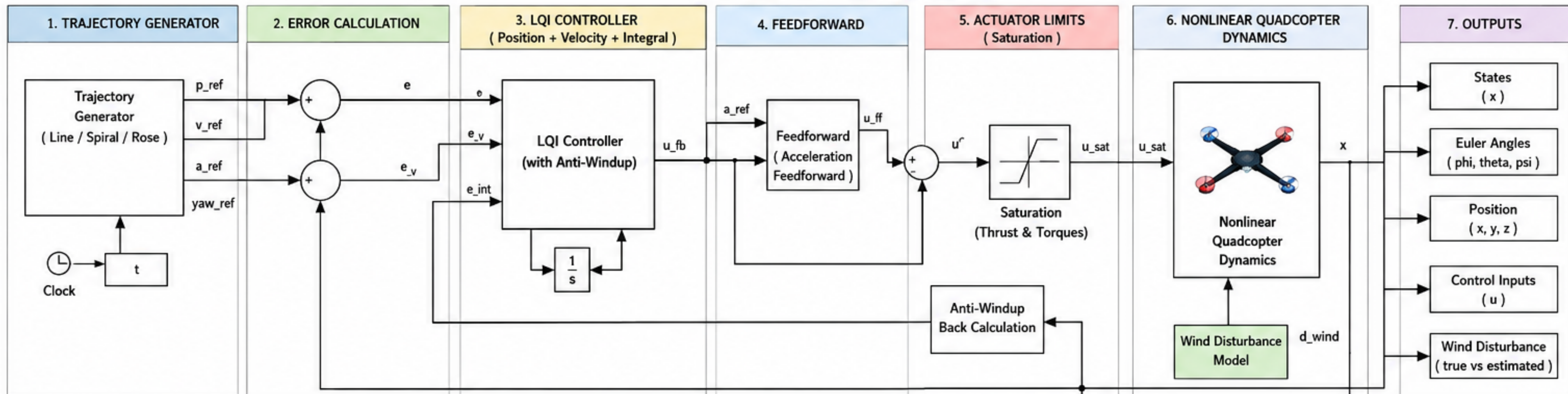
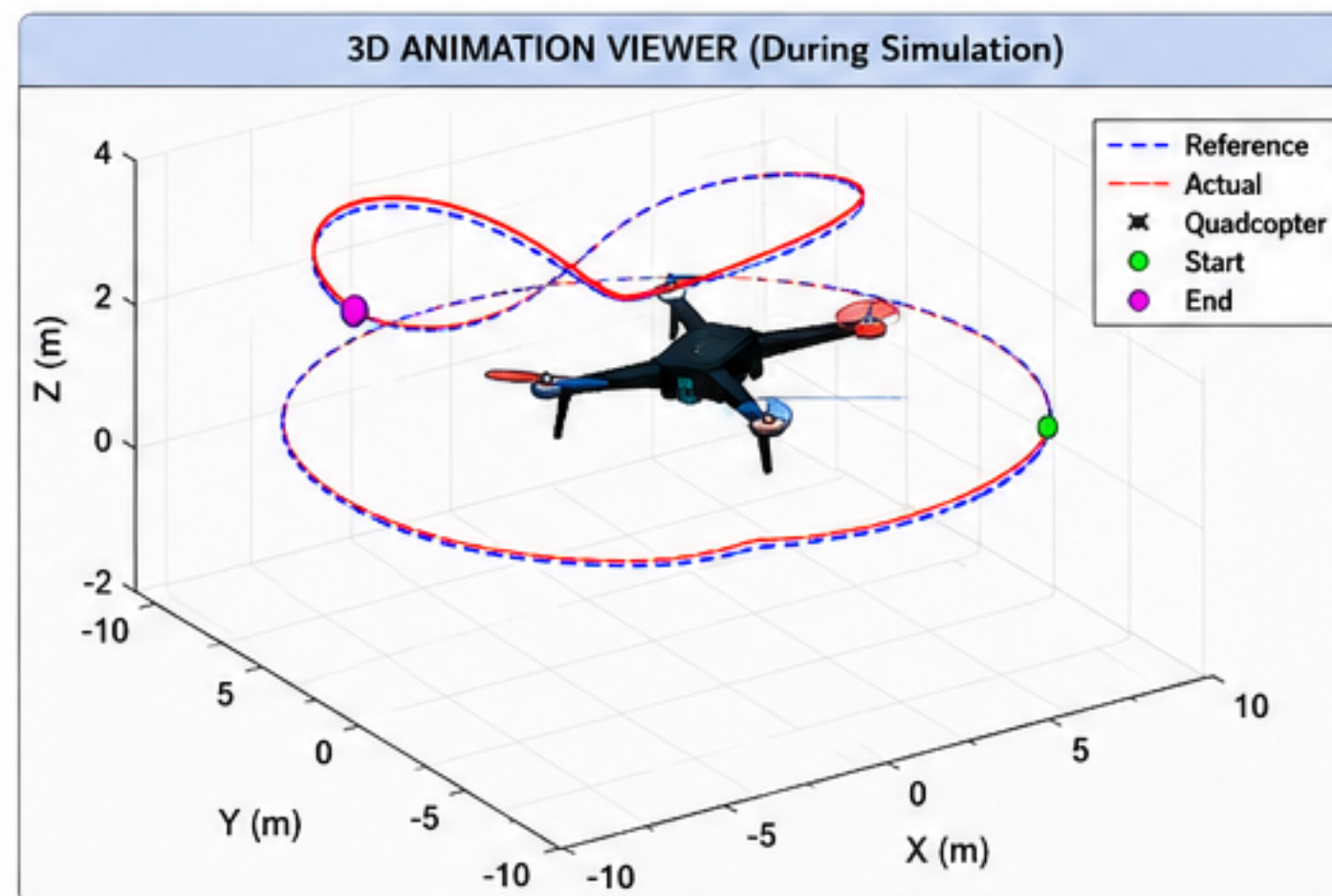
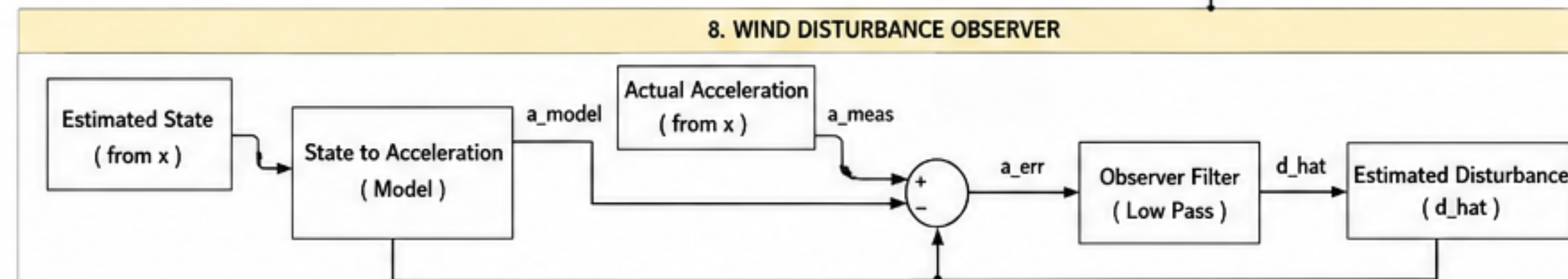


Quadcopter Nonlinear Model with LQI Controller, Feedforward, Anti-Windup, Wind Disturbance Observer and 3D Animation Viewer



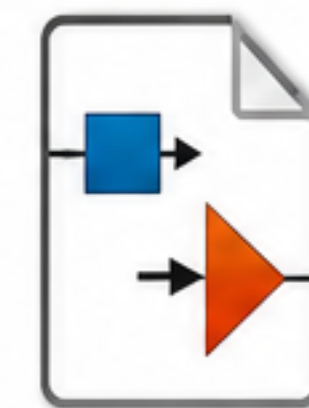
SIMULINK MODEL OVERVIEW

- Trajectories: Line / Spiral / Rose
- LQI Controller with Anti-Windup
- Acceleration Feedforward
- Actuator Saturation (Thrust & Torques)
- Nonlinear Quadcopter Dynamics
- Wind Disturbance Model
- Wind Disturbance Observer
- 3D Animation Viewer (MATLAB Figure)
- Logging & Performance Plots (RMSE, etc.)



HOW TO RUN

1. Open the model file `run_quadcopter_lqi_simulink.slx` in MATLAB/Simulink
2. Click the "Run" button (or press Ctrl + T)
3. 3D Animation window will open automatically
4. Check scopes and plots after simulation



run_quadcopter_lqi_simulink.slx
Simulink Model File